ADITI PANCHAL

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EDUCATION

Georgia Institute of Technology, Atlanta, GA — **Expected Dec 2027**
Degree: M.S. in Computer Science, Machine Learning Specialization
Cumulative GPA: 4.0/4.0

Texas A&M University, College Station, TX — **Dec 2021**
Degree: B.S. in Computer Science, Honors; *Minors:* Statistics, Psychology
Cumulative GPA: 3.7/4.0, *Major GPA:* 3.8/4.0

SKILLS AND ACCOMPLISHMENTS

Languages: Python, Java, C++, C#, SQL
ML: PyTorch, scikit-learn, XGBoost, PySpark, LLM evaluation, supervised learning, ranking evaluation, metric development, experimentation
ML/Cloud Platforms: Azure ML, Azure, AWS, Google Cloud
Publications: Closed Captioning for In-Person Lectures to Assist Students with Learning Disabilities (TAMU Undergraduate Research); Using GANs to Generate Random Surface Materials for High Fidelity Low Altitude Imaging Simulations (2020 IEEE Aerospace Conference)
Awards: Dean's Honor Roll, Undergraduate Research Scholar, Engineering Honors

EXPERIENCE

Microsoft | Redmond, WA — **May 2024 - Oct 2025**
Applied Scientist, Experiences + Devices, Copilot Experimentation + Metrics
- Adapted a Microsoft Research publication to design, train, and productionize a hybrid classifier (logistic regression on telemetry + LLM-extracted features) for user satisfaction prediction (~0.8 precision, ~0.7 recall); shipped as M365 Copilot's primary evaluation metric for executive and board-level reviews, and as a holistic quality-tracking metric across feature teams.
- Built evaluation models for Copilot (multi-turn intent/scenario classification, citation relevance, NDCG for retrieval ranking, and task-completion rate) using classical scoring and LLM-as-judge methods calibrated to match human-judge quality; engineered and scaled into automated pipelines used by every Enterprise Copilot team for online/offline experimentation and flight review.
- Compared evaluation models against human judgments using correlation and inter-rater agreement analyses, to improve alignment.
- Resolved 50+ incidents for metrics and/or experimentation pipelines as part of the engineering on-call rotation.

Technologies: Azure ML, Python, PySpark, PyTorch, xgboost, scikit-learn, CI/CD

Microsoft | Redmond, WA — **Mar 2022 - May 2024**
Software Engineer, Experiences + Devices, Search Relevance Data + Measurement
- Designed and shipped an LLM-based side-by-side judge that replaced human judges in Copilot's flight review process, achieving the highest correlation to human judgment among candidate metrics (Kendall's Tau 0.45); methodology approved via internal RFC for flight reviews.
- Benchmarked LLMs on quality/latency tradeoffs for production labeling pipeline; recommendation adopted org-wide and referenced by other teams for their own model selection.
- Developed scalable and customizable components for debugging production flight review pipelines for Experimentation Platform (for A/B testing) and visualizations for 15+ Search and Copilot teams to self-serve, in an eyes-on environment.

Technologies: Azure ML, Python, PySpark, CI/CD

Microsoft | Remote (original: Redmond, WA) — **May 2021 - Aug 2021**
Software Engineer Intern, Cloud + AI, Lead to Commerce

Capital One | Remote (original: Richmond, VA) — **Jun 2020 - Aug 2020**
Software Engineer Intern, Commercial Technology

RESEARCH AND PROJECTS

Vigil
Real-time Glucose Forecasting and Alerting System for Diabetes, Personal Project — **Sep 2026**
- Built an end-to-end alerting system and designed/evaluated 5 forecasting models (e.g. Kalman filter and a derived insulin decay model).
- In-progress: fine-tuning public time-series foundation models (Chronos) on benchmark CGM data; agentic checks for model regression.

Technologies: Python, TypeScript, Kotlin, Google Cloud, Android Studio, ESP32 hardware, Hugging Face

Sketch Recognition Lab | College Station, TX — **Aug 2020 - Dec 2021**
Research Scholar, Undergraduate Research Scholars Program
- Researched leveraging NLP for educational accessibility and developed an app that generates closed captions for in-person class lectures.

Technologies: JavaScript (React Native), Google Cloud

SpaceCRAFT Research Lab | College Station, TX — **Aug 2019 - May 2020**
Research Assistant, Machine Learning Team
- Co-authored IEEE 2020 publication on GAN-based simulation generation. Implemented Deep Q Learning for spacecraft auto-navigation.

Technologies: Python, Pandas, TensorFlow - Keras, OpenCV, NumPy